

1. Check all that apply to logical address translation.
The memory management unit may be inside the CPU.
Different CPU may have different translation schemes.
2. This exercise is about address translation on an x86 Intel CPU. This CPU translates (using hardware) a logical address LA into a physical address PA as follows:

```
if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP
```

Suppose DS = 0x2000 and CS = 0x0000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address 7000h

```
MOV SI, 3000h
MOV BX, [SI]
```

The physical address of the memory access instruction is _____ and the physical address of its memory operand is _____.

07000h
23000h

3. 4 processes are running on a system. There (is) are _____ independent logical space(s).
4
4. Suppose that a system performs address binding at execution time. Check all that applies.
After loading, a program can be moved around the physical space and still execute correctly.
The exact same binary program can execute correctly in different locations in the physical space.
This system offers great flexibility for the operating system to place programs
Every program can correctly work wherever it is loaded

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if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP
```

Suppose DS = 0x0000 and CS = 0x0000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address 0006h

```
MOV SI, E002h
MOV BX, [SI]
```

The physical address of the memory access instruction is _____ and the physical address of its memory operand is _____.

0006h
E002h

6. This exercise is about address translation on an x86 Intel CPU. This CPU translates (using hardware) a logical address LA into a physical address PA as follows:

```

if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP

```

Suppose DS = 0x0000 and CS = 0x0000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address 0008h

```

MOV SI, 0000h
MOV BX, [SI]

```

The physical address of the memory access instruction is _____ and the physical address of its memory operand is _____.

0008h

0000h

7. This exercise is about address translation on an x86 Intel CPU. This CPU translates (using hardware) a logical address LA into a physical address PA as follows:

```

if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP

```

Suppose DS = 0x0000 and CS = 0x0000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address 0004h

```

MOV SI, 2000h
MOV BX, [SI]

```

The physical address of the memory access instruction is _____ and the physical address of its memory operand is _____.

0004h

2000h

8. This exercise is about address translation on an x86 Intel CPU. This CPU translates (using hardware) a logical address LA into a physical address PA as follows:

```

if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP

```

Suppose DS = 0x2000 and CS = 0x5000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address 0600h

```

MOV DH, [1000h]

```

The physical address of the memory access instruction is _____ and the physical address of the operand [1000h] is _____.

50600h

21000h

9. On modern systems, address binding is performed at _____.

running time

execution time

10. This exercise is about address translation on an x86 Intel CPU. This CPU translates (using hardware) a logical address LA into a physical address PA as follows:

```
if (LA references data)
    PA = 10h*DS + LA
else // LA references an instruction, i.e. LA is in Register
    IP: the instruction pointer
    PA = 10h*CS + IP
```

Suppose DS = 0x0000 and CS = 0x0000. This question is about the instruction I that accesses data in memory. Let this memory access instruction I be at address F000h

```
MOV SI, 40A0h
MOV BX, [SI]
```

The physical address of the memory access instruction is _____ and the physical address of its memory operand is _____.

F000h
40A0h